Algorithm

An **algorithm** is a finite sequence of instructions to be followed for solving a problem.

An algorithm must have the following **properties**:

Input :

An algorithm has zero or more inputs. If an algorithm says to take inputs, it must receive some input data supplied externally.

Output:

An algorithm must produce at least one output as the result.

Finiteness:

No matter what is the input, the algorithm must terminate after a finite number of steps.

Definiteness:

The steps to be performed in the algorithm must be clear and unambiguous.

Effectiveness:

An algorithm must be developed by using very basic and simple operations so that one can trace it out by using just pen and paper.

Flowchart

A flowchart is a diagrammatic representation of an algorithm. It is a picture of the sequence of steps needed to solve a problem.

Computer Program

A computer program is a sequence of instructions *written in a programming language* that can be executed by a computer to perform a specific task (i.e. to solve a specific problem).

Symbols used in a Flowchart

Symbol Name	Symbol	function
Oval		Used to represent start and end of flowchart
Parallelogram		Used for input and output operation
Rectangle		Processing: Used for arithmetic operations and data-manipulations
Diamond		Decision making. Used to represent the operation in which there are two/three alternatives, true and false etc
Arrows		Flow line Used to indicate the flow of logic by connecting symbols